

EXPERIMENT 6

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Student Name: Pranav Sood

UID: 24BDA70188

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

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Subject Name: ADBMS

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Experiment 6 -- To study and implement advanced SQL queries using Joins, Corelated Query's and Conditional statement.

Aim : To study and implement correlated subqueries in SQL by retrieving employees whose salaries are among the top three highest distinct salary values within their respective departments, while correctly handling duplicate salary values using aggregate functions and joins

Question :6 (i) Calculate Apple and Orange Daily Sale difference

SQL Query:

```
SELECT sale_date, SUM(CASE WHEN fruit = 'apples' THEN sold_num ELSE -sold_num END) as s
FROM Sales
GROUP BY sale_date
```

	sale_date date	s bigint
1	2020-05-04	-1
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-01	2

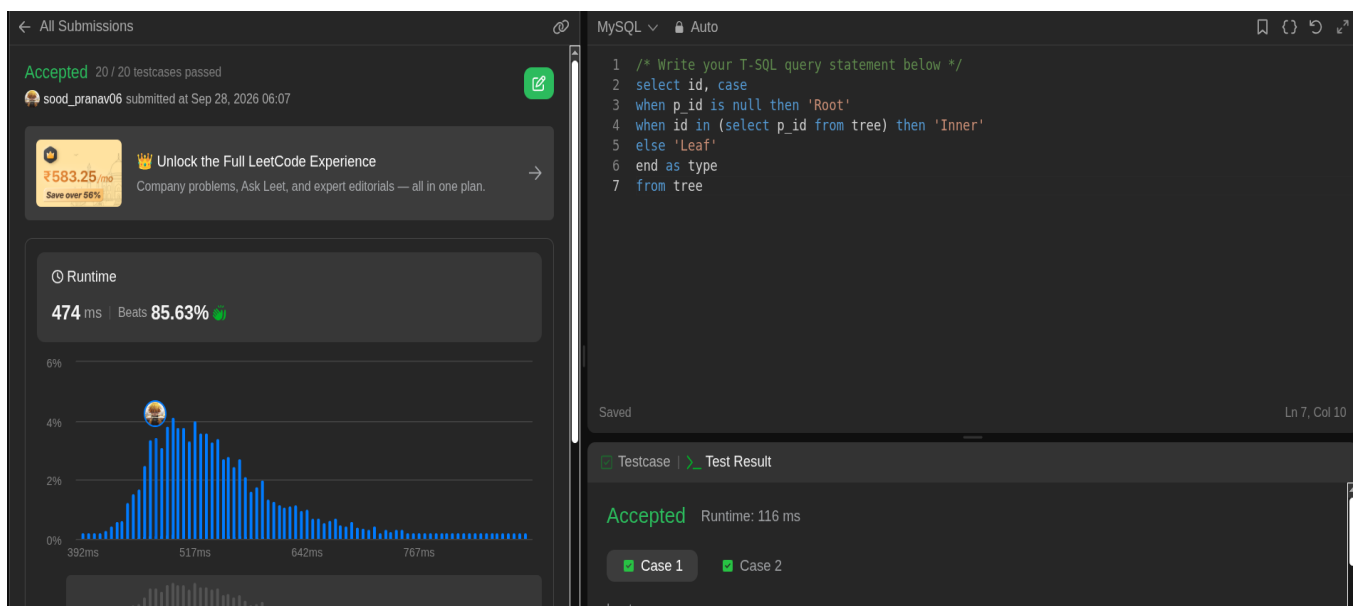
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Question: 6 (ii) Determine the type of Node

SQL Query:

```
SELECT DISTINCT t.id, (  
    CASE  
        WHEN t.p_id is NULL THEN 'Root'  
        WHEN l.id is NULL THEN 'Leaf'  
        ELSE 'Inner'  
    END  
)as type  
FROM Tree as t  
LEFT JOIN Tree as l  
ON t.id = l.p_id
```



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Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand and implement correlated subqueries** to solve SQL problems involving row-wise comparisons.
2. **Apply aggregate functions** such as COUNT() and DISTINCT to retrieve top-N records within groups.
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.
4. **Analyze and solve complex SQL queries** involving grouped data, ranking logic, and handling duplicate values effectively.